

THE ELEMENTARY AND
PRINCIPLES OF FINANCE



KUTUSH KUMAR, KUTUM PAKHI

Variable input, 82–83

w

Wage, 90–91, 96, 112–114, 117–120

Want, 2–4, 20, 32, 47–48, 50

Willingness to pay, 51

Long-run average cost curve, 93

M

Macroeconomics, 14–16

Marginal cost, 90, 92, 95–98, 100–107

Marginal product, 80–87, 90, 97, 116

Marginal productivity theory, 116

Marginal rate of substitution, 52

Marginal revenue, 100, 102, 106–107

Marginal utility, 20–27

Market, 3–4, 6–7, 14–15, 17, 21–23, 26, 31–33, 37, 41–43, 47, ...

Market demand, 32

Market structure, 100, 105, 108

Microeconomics, 12, 14–16, 18

Model, 14

Monopolistic competition, 100, 103–104, 107–108

Monopoly, 100, 102–104, 107–109

N

Nominal wage, 112–114, 118

Normal good, 33

Normal profit, 112, 115, 118

Normative statement, 13

O

Oligopoly, 105

Opportunity cost, 2, 4–5, 7–8

Ordinal measurement, 23

Output, 3, 13, 15–16, 21, 31, 35, 41, 51, 61, 64, ...

P

Percentage change, 41–42, 44, 62, 122–123

Perfect competition, 100–105, 107–108

Perfectly elastic demand, 43

Perfectly inelastic demand, 43

Positive statement, 13

Price ceiling, 72, 77–78

Price elasticity of demand, 41

Price elasticity of supply, 62

Price floor, 72

Price maker, 102

Price taker, 101

Prime cost, 90–94, 96

Producer surplus, 70, 73–74, 76

INDEX

A

Accounting profit, 112, 115, 118–120
Art, 12–13
Average cost, 90, 92–93, 95–98, 100, 107
Average product, 80–85, 87

B

Backward-bending supply curve, 62
Bandwagon effect, 30, 34, 36
Bilateral monopoly, 104
Budget line, 50–52, 54–57

C

Cardinal measurement, 22
Ceteris paribus, 14
Choice, 2, 4–5, 7–8, 12, 14–15, 20, 30, 40, 50, 56, ...
Competitive market, 101
Complementary goods, 33
Constant returns to scale, 84
Consumer, 20, 50–53, 56–57, 70, 73–74, 76
Consumer equilibrium, 52–53
Consumer surplus, 50–53, 56–57, 70, 73, 76
Consumption, 50–51
Cost, 2–5, 7–9, 12–13, 16–17, 20–21, 26, 30–31, 35, 37, 40–41, ...
Cross elasticity of demand, 44

D

Decreasing returns to scale, 85
Demand, 20, 26, 30–35, 37–38, 41–45, 47, 53, 70–77, 106, 117
Demand curve, 31, 33–35, 45, 53
Demand schedule, 31
Determinants of demand, 32
Determinants of supply, 62
Differential rent, 112–120

E

Economic law, 12–14, 18
Economic organisation, 5–6
Economic profit, 112, 115, 118–120
Economic rent, 112
Economic system, 6
Economics, 2, 6, 8, 16, 47
Elastic demand, 42–43

10.4 | Least-Cost Combination | technique | 550 94

XI. MARKET STRUCTURES AND THE FIRM 99

11.1 | Types of Market | framework | 225 100
 11.2 | Perfect Competition | new idea | 775 101
 11.3 | Monopoly | new idea | 550 102
 11.4 | Monopolistic Competition | new idea | 550 103
 11.5 | Pricing and the Firm's Goal | framework | 225 104

XII. DISTRIBUTION: RENT, WAGES, INTEREST AND PROFIT 111

12.1 | Rent | defining theory | 225 112
 12.2 | Wages | framework | 775 113
 12.3 | Interest | framework | 225 114
 12.4 | Profit | framework | 775 116

APPENDIX A. MATHS YOU WILL USE 121

A.1 A1.1 | Percentages and percentage change | technique | 250 122
 A.2 A1.2 | Ratios, averages and rounding | technique | 250 122
 A.3 A1.3 | Rearranging an equation | technique | 250 122
 A.4 A1.4 | Reading a graph | technique | 250 123

GLOSSARY 125**ANSWERS TO QUESTIONS 137****FURTHER READING 143****INDEX 145**

FURTHER READING

Alfred Marshall introduces elasticity of demand in Principles of Economics, 1890, historical case reference

David Ricardo's rent theory in his 1815 essay and his 1817 Principles, and the Corn Laws repealed in 1846, historical case reference

Jules Dupuit, 1844, on measuring the usefulness of public works such as bridge tolls, historical case reference

The marginal revolution of 1871 to 1874: Jevons, Menger and Walras explain the diamond-and-water puzzle, historical case reference

US petrol shortage after the October 1973 oil embargo, with queues and the odd-even plate rationing of 1974, historical case reference

CHAPTER IX · PRODUCTION

9.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

9.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

9.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

9.4 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

9.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

9.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

9.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER X · COSTS OF PRODUCTION

10.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

10.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

10.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

10.4 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

10.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

10.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

10.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Show why every society and every shop must choose, and give the reader the first working words of economics. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet; two paid assistants. Record of 12 January 2026: restock fund Tk 400,000 in cash from family savings; shelf space allows one large restock only. Option A, textbooks: purchase cost Tk 400,000; expected sales Tk 560,000; expected sales less purchase cost Tk 160,000. Option B, stationery: purchase cost Tk 400,000; expected sales Tk 540,000; expected sales less purchase cost Tk 140,000. Decision recorded: option A. Opportunity cost of that decision: Tk 140,000. The three problems in shop terms: what to stock (textbooks), how to run the shop (family at the counter plus two assistants), for whom (university students first, schools second). The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

**1.1 | WANTS, RESOURCES AND SCARCITY | NEW
IDEA | 775**

This section asks what wants, resources and scarcity means before applying it.

Economics. the study of how people, firms and societies choose among uses of limited resources to meet wants

Economics is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Economics** means the study of how people, firms and societies choose among uses of limited resources to meet wants. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wants, resources and scarcity, before moving to the next step.

CHAPTER V · ELASTICITY OF DEMAND

5.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

5.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

5.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

5.4 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

5.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

5.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

5.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER VI · CONSUMER CHOICE: SURPLUS AND INDIFFERENCE CURVES

6.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

6.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

6.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

6.4 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

6.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

6.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

6.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

1.2 | CHOICE AND OPPORTUNITY COST | NEW IDEA | 550

This section asks what choice and opportunity cost means before applying it.

Good. a physical thing people buy to satisfy a want, such as a book or a pen

Good is the working idea for this part of the chapter.

Service. an activity done for someone, such as binding a thesis or delivering parcels

Service is the working idea for this part of the chapter.

Scarcity. the condition in which resources are too few to satisfy all the wants people have

Scarcity is the working idea for this part of the chapter.

Choice. picking one option and giving up the others because resources cannot cover every option

Choice is the working idea for this part of the chapter.

Choice and Opportunity Cost is understood by separating the question into small steps.

Choice and Opportunity Cost is understood by separating the question into small steps.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Good** means a physical thing people buy to satisfy a want, such as a book or a pen. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in choice and opportunity cost, before moving to the next step.

The term **Service** means an activity done for someone, such as binding a thesis or delivering parcels. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in choice and opportunity cost, before moving to the next step.

The term **Scarcity** means the condition in which resources are too few to satisfy all the wants people have. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in choice and opportunity cost, before moving to the next step.

The term **Choice** means picking one option and giving up the others because resources cannot cover every option. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in choice and opportunity cost, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because

ANSWERS TO QUESTIONS

CHAPTER I · SCARCITY, CHOICE AND THE ECONOMIC PROBLEM

1.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

1.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

1.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

1.4 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

1.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

1.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

1.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER II · ECONOMIC REASONING, LAWS AND THE SCOPE OF MICROECONOMICS

2.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

2.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

2.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

2.4 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

2.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

2.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

2.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

The Basic Problems of an Economic Organisation is understood by separating the question into small steps.

The Basic Problems of an Economic Organisation is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what defining economics means before applying it.

Economic system. the set of rules and institutions a country uses to answer its basic economic problems

Economic system is the working idea for this part of the chapter.

Defining Economics is understood by separating the question into small steps.

Defining Economics is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

precisely what is being measured or compared in the basic problems of an economic organisation, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in the basic problems of an economic organisation to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

1.4 | DEFINING ECONOMICS | DEFINING THEORY | 225

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Economic system** means the set of rules and institutions a country uses to answer its basic economic problems. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in defining economics, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in defining economics is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in defining economics to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion

shows how the term works before it is applied to a changed situation. (*Chapter V*)

Total surplus. consumer surplus plus producer surplus, a measure of the gain from trade. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Total utility. the whole satisfaction from all the units of a good consumed in a period. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Unit elastic demand. demand for which the percentage change in quantity equals the percentage change in price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Util. an imaginary unit used to count satisfaction in examples. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Utility. the satisfaction a person gets from using a good or service. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Variable cost. a cost that rises and falls with the amount produced. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Variable input. an input whose quantity can be changed even in the short run. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Wage. the payment made to labour for its work. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Want. something a person would like to have, whether or not they can pay for it. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Willingness to pay. the highest price a buyer would accept to get one unit of a good. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VI*)

return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Show why every society and every shop must choose, and give the reader the first working words of economics. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

- 1.T1 scarcity versus shortage, forgone value of a choice made from a new price list, which of the three problems a given decision belongs to
- 1.T2 a student has Tk 9,000 and one free Saturday and must pick between two paid tasks and a course fee — name the opportunity cost

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to US petrol shortage after the October 1973 oil embargo, with queues and the odd-even plate rationing of 1974; search: 1973 oil embargo gasoline lines odd-even rationing. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

SOURCES

Further reading and historical attribution should be checked against the original works and reliable archival or institutional sources. The chapter does not require the reader to accept a historical interpretation merely because it appears in a secondary account.

REVIEW QUESTIONS

- 1.3 scarcity versus shortage, forgone value of a choice made from a new price list, which of the three problems a given decision belongs to
- 1.4 a student has Tk 9,000 and one free Saturday and must pick between two paid tasks and a course fee — name the opportunity cost

The chapter has built one connected chain of reasoning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

The chapter uses a small number of historical references to anchor its defining ideas.

Production. the process of turning resources into goods and services. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Production function. a rule or table linking the quantities of inputs used to the greatest output they can make. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Profit. the reward of enterprise: what remains of revenue after the other factors are paid. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Profit maximisation. choosing the output at which the gap between total revenue and total cost is greatest. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Pure competition. a market with many sellers, an identical product and free entry, where buyers may lack full information or free movement. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Pure monopoly. a monopoly in which one seller supplies the whole market with a product that has no close substitute. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Quantity demanded. the amount people plan to buy at one particular price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Quantity supplied. the amount sellers plan to offer at one particular price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Rate of interest. the interest paid in a year as a percentage of the amount borrowed. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Real wage. the amount of goods and services a nominal wage can buy. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Resource. anything used to make goods and services: land, labour, capital and enterprise. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Returns to scale. the change in output when all inputs are changed by the same proportion. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Risk bearing. taking on the chance of loss when production is uncertain. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Saturation point. the quantity at which marginal utility reaches zero and total utility stops rising. The chapter uses this definition consistently; the running

Microeconomics. the study of choices made by single households, firms and markets. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Model. a simplified picture of a situation that keeps only the features needed to answer a question. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Monopolistic competition. a market with many sellers of similar but not identical goods and free entry. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Monopoly. a market with one seller and no close substitute, so that seller sets the price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Nominal wage. the wage counted in money. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Normal good. a good for which demand rises when buyers' incomes rise. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Normal profit. the least profit that keeps a firm in business, equal to the implicit costs of the owner's own resources. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Normative statement. a claim about what ought to be, resting on values and opinion. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Oligopoly. a market with a few large sellers, each aware of the others' actions. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Opportunity cost. the value of the best alternative given up when a choice is made. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Ordinal measurement. ranking satisfaction as first, second and third without giving it a number. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Output. the goods or services produced. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Percentage change. the size of a change divided by the starting value, multiplied by 100. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Teach the reader how economists argue, what an economic law is, and where microeconomics fits. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at Tk 60 and 200 copies sell in a week. Ledger note A, in Shirin Meghna's hand: when the price was raised to Tk 66 on 2 March 2026, weekly sales fell to 188. Ledger note B, in Imran Meghna's hand: students should not have to pay more than Tk 60. Note A can be checked against sales records; note B cannot. Regular tendency recorded over three years: textbook sales rise in the two weeks before each term begins (this holds when the shop's prices stay unchanged). Micro and macro: the shop's price and stock are micro matters; the movement of the average price of all goods in Bangladesh is a macro matter. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

2.1 | SCIENCE, ART AND NORMATIVE VIEWS | NEW IDEA | 550

This section asks what science, art and normative views means before applying it.

Science. a body of knowledge built by observing facts, forming explanations and testing them

Science is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Science** means a body of knowledge built by observing facts, forming explanations and testing them. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in science, art and normative views, before moving to the next step.

Indifference curve. a line joining bundles of two goods that give a consumer the same satisfaction. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (Chapter VI)

Indifference map. a set of indifference curves, with curves farther from the origin showing more satisfaction. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (Chapter VI)

Industry. all the firms that make the same or very similar goods. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (Chapter XI)

Inelastic demand. demand for which the percentage fall in quantity is smaller than the percentage rise in price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (Chapter V)

Inelastic supply. supply for which quantity responds by a smaller percentage than price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (Chapter VII)

Inferior good. a good for which demand falls when buyers' incomes rise. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (Chapter IV)

Input. a resource used in production. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (Chapter IX)

Interest. the payment for the use of borrowed money or capital over time. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (Chapter XII)

Isocost line. a line joining the input combinations that all cost the same total. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (Chapter X)

Isoquant. a curve joining the input combinations that all make the same output. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (Chapter X)

Law of demand. when the price of a good rises, quantity demanded falls, and when it falls, quantity demanded rises, other things equal. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (Chapter IV)

Law of diminishing marginal utility. as a person consumes more units of a good, the extra satisfaction from each further unit falls. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (Chapter III)

Law of diminishing returns. when more of one input is added to fixed inputs, the extra output eventually falls. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (Chapter IX)

Law of supply. when the price of a good rises, quantity supplied rises, and when it falls, quantity supplied falls, other things equal. The chapter uses this

Ceteris paribus. a Latin phrase meaning other things being equal, used to study one change at a time

Ceteris paribus is the working idea for this part of the chapter.

Economic law. a statement of a regular tendency in human choices, true when other things stay equal

Economic law is the working idea for this part of the chapter.

Economic Laws is understood by separating the question into small steps.

Economic Laws is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what microeconomics and macroeconomics means before applying it.

Model. a simplified picture of a situation that keeps only the features needed to answer a question

Model is the working idea for this part of the chapter.

The term **Ceteris paribus** means a Latin phrase meaning other things being equal, used to study one change at a time. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in economic laws, before moving to the next step.

The term **Economic law** means a statement of a regular tendency in human choices, true when other things stay equal. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in economic laws, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in economic laws to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

2.3 | MICROECONOMICS AND MACROECONOMICS | FRAMEWORK | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Model** means a simplified picture of a situation that keeps only the features needed to answer a question. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or

shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Economic law. a statement of a regular tendency in human choices, true when other things stay equal. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Economic organisation. the way a society or business arranges who decides what is made, how and for whom. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Economic profit. revenue less all costs, explicit and implicit. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Economic rent. the payment to a factor above the minimum needed to keep it in its present use. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Economic system. the set of rules and institutions a country uses to answer its basic economic problems. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Economics. the study of how people, firms and societies choose among uses of limited resources to meet wants. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Elastic demand. demand for which the percentage fall in quantity is greater than the percentage rise in price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Elastic supply. supply for which quantity responds by a larger percentage than price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Elasticity. a measure of how strongly one quantity responds when another quantity changes. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Equilibrium. a state in which nothing pushes the price or quantity to change. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Equilibrium price. the price at which quantity demanded equals quantity supplied. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Equilibrium quantity. the quantity bought and sold at the equilibrium price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Excess demand. the amount by which quantity demanded exceeds quantity supplied at a price below equilibrium. The chapter uses this definition consistently; the running example

Scope, Importance and Objectives is understood by separating the question into small steps.

Scope, Importance and Objectives is understood by separating the question into small steps.

Scope, Importance and Objectives is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in scope, importance and objectives is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in scope, importance and objectives to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

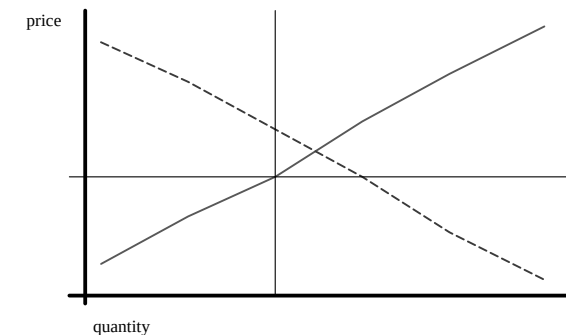


FIG. 2.1. a two-branch map of economics showing the questions that belong to microeconomics and to macroeconomics

The same method remains useful when the surrounding details change.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

GLOSSARY

Accounting profit. revenue less explicit costs only. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. *(Chapter XII)*

Art. the skilled use of knowledge to reach a practical aim. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. *(Chapter II)*

Average cost. total cost divided by the number of units produced. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. *(Chapter X)*

Average product. total product divided by the number of units of the variable input. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. *(Chapter IX)*

Backward-bending supply curve. a curve showing that beyond some pay level a worker offers fewer hours as pay rises further. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. *(Chapter VII)*

Bandwagon effect. a rise in demand for a good because more people are seen to buy it. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. *(Chapter IV)*

Bilateral monopoly. a market with a single seller facing a single buyer. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. *(Chapter XI)*

Budget line. a line showing every bundle of two goods a consumer can exactly afford with a given income. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. *(Chapter VI)*

Cardinal measurement. measuring satisfaction in numbers, such as 20 utils, so amounts can be added and compared. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. *(Chapter III)*

Ceteris paribus. a Latin phrase meaning other things being equal, used to study one change at a time. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. *(Chapter II)*

Choice. picking one option and giving up the others because resources cannot cover every option. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. *(Chapter I)*

Competitive market. a market with many buyers and sellers, so no single one can set the price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. *(Chapter XI)*

Complementary goods. goods used together, so a rise in one's price lowers demand for the other. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. *(Chapter IV)*

SUMMARY

Teach the reader how economists argue, what an economic law is, and where microeconomics fits. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

2.T1 sort five statements into positive and normative, state a law with its other-things-equal condition, place four decisions under micro or macro

2.T2 a minister says bread should be cheaper — separate the checkable part from the opinion

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to Great Debasement under Henry VIII, 1544 to 1551, and Thomas Gresham's observation about money; search: Great Debasement Gresham's law. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

REVIEW QUESTIONS

2.3 sort five statements into positive and normative, state a law with its other-things-equal condition, place four decisions under micro or macro

2.4 a minister says bread should be cheaper — separate the checkable part from the opinion

2.5 after the other-things-equal passage and after the micro-macro split

PAUSE AND CHECK 2.P6 after the other-things-equal passage

PAUSE AND CHECK 2.P7 the micro-macro split

The chapter has built one connected chain of reasoning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

A.4 A1.4 | READING A GRAPH | TECHNIQUE | 250

In reading a graph, write the given numbers before choosing an operation. Keep the units visible, carry the arithmetic line by line, and round only at the point required by the question. For a percentage change, compare the change with the starting value. For an average, add the values and divide by their count. For an equation, perform the same operation on both sides. For a graph, read the axes before reading a point.

Given a price of Tk 60 rising to Tk 66, the change is Tk 6. The percentage change is $6 \div 60 \times 100 = 10$ per cent. The important check is that the starting value is 60, not 66. The same discipline applies to ratios, equations and graphs: identify the reference quantity before calculating.

Reading a graph is a tool for checking numerical relationships.

A tiny worked example makes the procedure concrete.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Explain satisfaction as the root of demand and give the first law the reader can test. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): total utility 20, 34, 43, 48, 50; marginal utility 20, 14, 9, 5, 2. A sixth notebook is rated 0 and a seventh would lower satisfaction by 3. Conditions noted on the card: all notebooks of the same kind, bought in a single visit, no change in taste. Notebook price Tk 60 (carried from the shop's price list). The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

3.1 | UTILITY AND SATISFACTION | NEW IDEA | 775

This section asks what utility and satisfaction means before applying it.

Utility. the satisfaction a person gets from using a good or service
Utility is the working idea for this part of the chapter.

Consumer. a person who buys or uses goods and services to meet wants

Consumer is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Utility** means the satisfaction a person gets from using a good or service. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in utility and satisfaction, before moving to the next step.

The term **Consumer** means a person who buys or uses goods and services to meet wants. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is

Util, an imaginary unit used to count satisfaction in examples
Util is the working idea for this part of the chapter.

Cardinal measurement.
measuring satisfaction in numbers, such as 20 utils, so amounts can be added and compared
Cardinal measurement is the working idea for this part of the chapter.

The Law of Diminishing Marginal Utility is understood by separating the question into small steps.

The Law of Diminishing Marginal Utility is understood by separating the question into small steps.

The Law of Diminishing Marginal Utility is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what limitations of the law means before applying it.

The term **Util** means an imaginary unit used to count satisfaction in examples. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of diminishing marginal utility, before moving to the next step.

The term **Cardinal measurement** means measuring satisfaction in numbers, such as 20 utils, so amounts can be added and compared. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of diminishing marginal utility, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in the law of diminishing marginal utility to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

3.3 | LIMITATIONS OF THE LAW | FRAMEWORK | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

Maths You Will Use



first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Explain how the money earned from production is shared among land, labour, capital and enterprise. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

12.T1 compute differential rents for new plots, compute a real wage from new price data, separate economic profit from accounting profit in a new account

12.T2 explain why the poorest plot in use earns no rent

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to David Ricardo's rent theory in his 1815 essay and his 1817 Principles, and the Corn Laws repealed in 1846; search: Ricardo rent Corn Laws 1815 repeal 1846. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

SOURCES

Further reading and historical attribution should be checked against the original works and reliable archival or institutional sources. The chapter does not require the reader to

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

The chapter has built one connected chain of reasoning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

The chapter uses a small number of historical references to anchor its defining ideas.

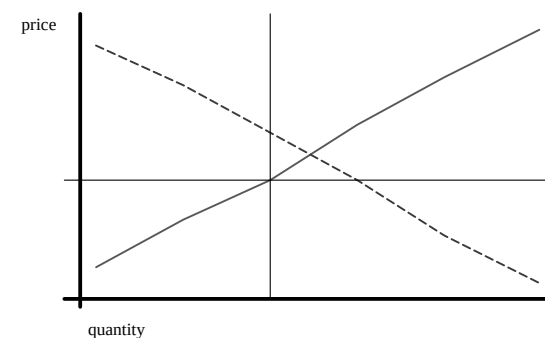


FIG. 3.1. the total utility curve rising and flattening against units consumed

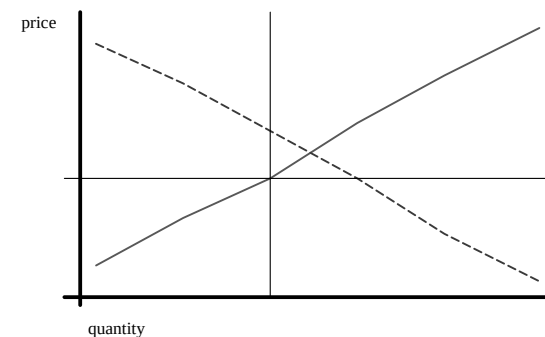


FIG. 3.2. the marginal utility curve falling to zero against the same units

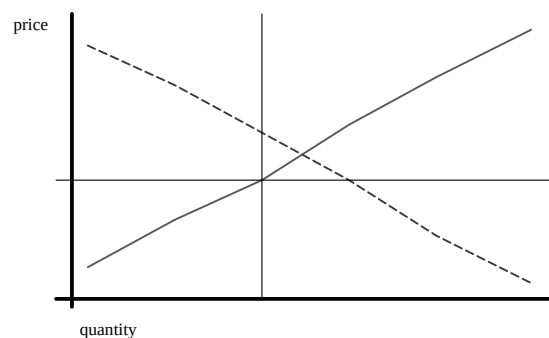


FIG. 12.1. differential rent on three plots of different yield

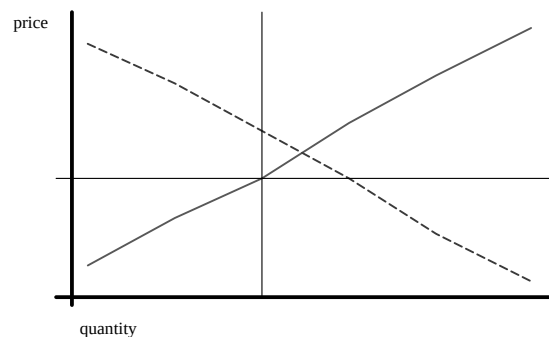


FIG. 12.2. a labour market with demand and supply setting the wage

The same method remains useful when the surrounding details change.

The worked case applies the chapter without adding facts to the record.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

The chapter has built one connected chain of reasoning.

rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): total utility 20, 34, 43, 48, 50; marginal utility 20, 14, 9, 5, 2. A sixth notebook is rated 0 and a seventh would lower satisfaction by 3. Conditions noted on the card: all notebooks of the same kind, bought in a single visit, no change in taste. Notebook price Tk 60 (carried from the shop's price list). Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Explain satisfaction as the root of demand and give the first law the reader can test. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

3.T1 complete a total-and-marginal table from new scores, state the law with its conditions, give one apparent exception and say why it is not one

reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Profit** means the reward of enterprise: what remains of revenue after the other factors are paid. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in interest, before moving to the next step.

The term **Economic profit** means revenue less all costs, explicit and implicit. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in interest, before moving to the next step.

The term **Accounting profit** means revenue less explicit costs only. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in interest, before moving to the next step.

The term **Normal profit** means the least profit that keeps a firm in business, equal to the implicit costs of the owner's own resources. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in interest, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in interest to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

Profit, the reward of enterprise: what remains of revenue after the other factors are paid

Profit is the working idea for this part of the chapter.

Economic profit, revenue less all costs, explicit and implicit

Economic profit is the working idea for this part of the chapter.

Accounting profit, revenue less explicit costs only

Accounting profit is the working idea for this part of the chapter.

Normal profit, the least profit that keeps a firm in business, equal to the implicit costs of the owner's own resources

Normal profit is the working idea for this part of the chapter.

Interest is understood by separating the question into small steps.

Interest is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in rent, before moving to the next step.

The term **Differential rent** means the extra return a better plot earns over the poorest plot in use. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in rent, before moving to the next step.

The term **Wage** means the payment made to labour for its work. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in rent, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

Before moving on, keep one question in view: what must remain true for the rule in rent to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

12.2 | WAGES | FRAMEWORK | 775

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Nominal wage** means the wage counted in money. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wages, before moving to the next step.

Differential rent, the extra return a better plot earns over the poorest plot in use

Differential rent is the working idea for this part of the chapter.

Wage, the payment made to labour for its work

Wage is the working idea for this part of the chapter.

Rent is understood by separating the question into small steps.

Rent is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what wages means before applying it.

Nominal wage, the wage counted in money

Nominal wage is the working idea for this part of the chapter.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Build the demand idea from a price table to a curve, and show what moves along it and what shifts it. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188; Tk 70, 180. Shock to determinants: schools announced a drawing term on 10 March 2026, and weekly sales at Tk 60 rose from 200 to 230 (a shift of the curve). Complements: when pen-pack price was cut from Tk 100 to Tk 90 on 16 March 2026, notebook sales at Tk 60 rose from 200 to 210 a week. Substitutes: when the price of a second-hand prescribed textbook fell by 20 per cent, weekly sales of the new copy fell from 50 to 44 (new copy price Tk 700). Snob effect: a numbered edition at Tk 2,400 sold out in three days, while the unnumbered Tk 1,200 edition took three weeks. Bandwagon effect: a title at Tk 450 sold 20 copies a week until it topped a public bestseller list, then 85 a week. Giffen good: no case in the shop; explain only in general terms. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

4.1 | WHAT DEMAND MEANS | NEW IDEA | 450

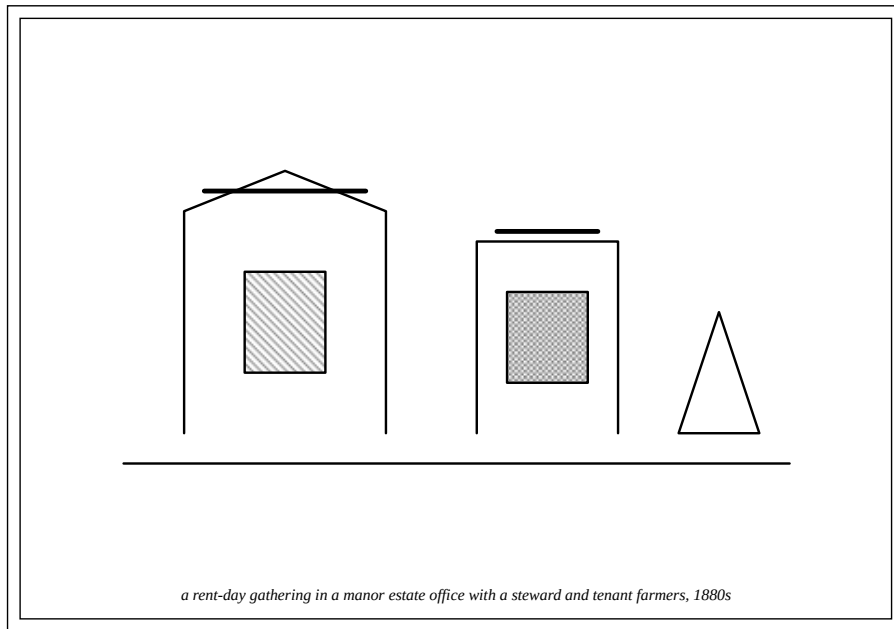
The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Demand** means the quantity of a good people are willing and able to buy at each possible price, in a given period. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what demand means, before moving to the next step.

This section asks what what demand means means before applying it.

Demand, the quantity of a good people are willing and able to buy at each possible price, in a given period

Demand is the working idea for this part of the chapter.



CHAPTER XII

Distribution: Rent, Wages, Interest and Profit

What does this chapter let us explain that the earlier ideas could not explain alone?



Explain how the money earned from production is shared among land, labour, capital and enterprise. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

Market. any arrangement that brings buyers and sellers of a good together

Market is the working idea for this part of the chapter.

Market demand. the sum of all individual buyers' demands for a good at each price

Market demand is the working idea for this part of the chapter.

Law of demand. when the price of a good rises, quantity demanded falls, and when it falls, quantity demanded rises, other things equal

Law of demand is the working idea for this part of the chapter.

Determinants of demand. the things other than the good's own price that change how much people want to buy

Determinants of demand is the working idea for this part of the chapter.

The Law of Demand is understood by separating the question into small steps.

The Law of Demand is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Market** means any arrangement that brings buyers and sellers of a good together. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of demand, before moving to the next step.

The term **Market demand** means the sum of all individual buyers' demands for a good at each price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of demand, before moving to the next step.

The term **Law of demand** means when the price of a good rises, quantity demanded falls, and when it falls, quantity demanded rises, other things equal. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of demand, before moving to the next step.

The term **Determinants of demand** means the things other than the good's own price that change how much people want to buy. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of demand, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in the law of demand to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

REVIEW QUESTIONS

11.3 find the profit-maximising output from a new cost table in each of three structures, name the structure from a list of features, decide whether a price cut helps revenue

11.4 assess the view that firms always aim to maximise profit

11.5 after the price-taker passage and after the monopoly marginal-revenue table

PAUSE AND CHECK 11.P6 after the price-taker passage

PAUSE AND CHECK 11.P7 the monopoly marginal-revenue table

The section closes by restating the condition that controls the result.

This section asks what related goods and exceptions means before applying it.

Giffen good. a rare inferior good for which a price rise leads buyers to buy more of it

Giffen good is the working idea for this part of the chapter.

Snob effect. a fall in demand for a good because more people begin to own it

Snob effect is the working idea for this part of the chapter.

Bandwagon effect. a rise in demand for a good because more people are seen to buy it

Bandwagon effect is the working idea for this part of the chapter.

Related Goods and Exceptions is understood by separating the question into small steps.

Related Goods and Exceptions is understood by separating the question into small steps.

condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in what moves the demand curve to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

4.4 | RELATED GOODS AND EXCEPTIONS | NEW IDEA | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Giffen good** means a rare inferior good for which a price rise leads buyers to buy more of it. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in related goods and exceptions, before moving to the next step.

The term **Snob effect** means a fall in demand for a good because more people begin to own it. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in related goods and exceptions, before moving to the next step.

The term **Bandwagon effect** means a rise in demand for a good because more people are seen to buy it. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in related goods and exceptions, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in related goods and exceptions is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The

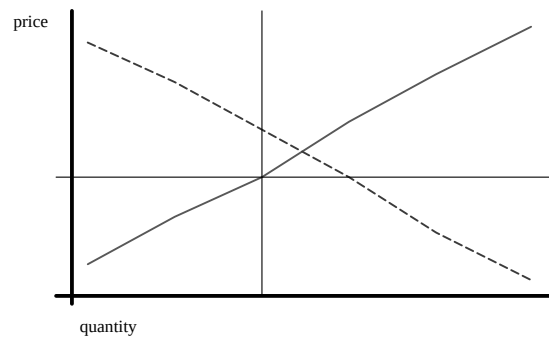


FIG. 11.4. a monopolistic competitor's short-run profit and long-run zero profit side by side

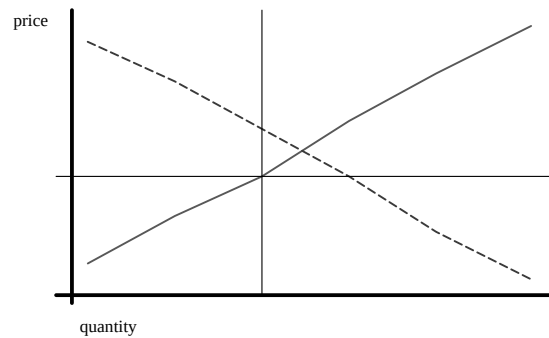


FIG. 11.5. a revenue comparison of a price cut for an elastic good and an inelastic good

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70; 50, Tk 80. Profit-maximising output 30 (marginal cost equals Tk 60). Average cost at 30 books Tk 52; profit $(60 - 52) \times 30 = \text{Tk } 240$ a day. Entry of new shops lowers price toward average cost in the long run. Monopoly: sole authorised seller of one foreign reference work. Price and weekly quantity: Tk 1,000, 10 copies; Tk 900, 20; Tk 800, 30; Tk 700, 40. Total revenue Tk 10,000; 18,000; 24,000; 28,000. Marginal revenue between steps Tk 800; 600; 400. Marginal cost constant Tk 500. Profit-maximising output 30 copies at Tk 800; profit $(800 - 500) \times 30 = \text{Tk } 9,000$. Monopolistic competition: a mystery novel series differentiated by cover and author; price Tk 400; short run average cost Tk 340 at 50 copies a week; profit Tk 3,000. After two more shops enter, the shop sells 40 copies at Tk 380 with average cost Tk 380; profit

The worked case applies the chapter without adding facts to the record.

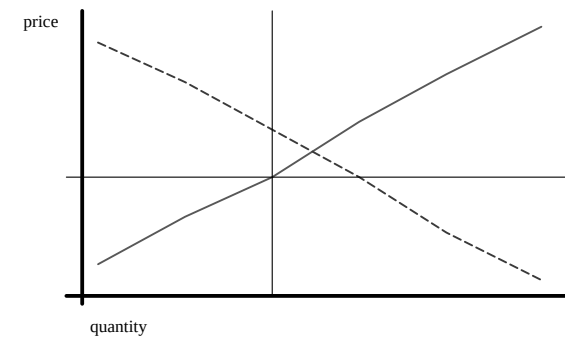


FIG. 4.3. a rightward and a leftward shift with the causes labelled

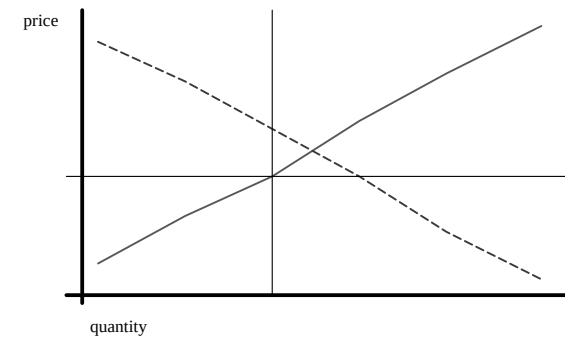


FIG. 4.4. the curve of an exceptional good that bends upward, with the snob and bandwagon cases beside it

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188; Tk 70, 180. Shock to determinants: schools announced a drawing term on 10 March 2026, and weekly sales at Tk 60 rose from 200 to 230 (a shift of the curve). Complements: when pen-pack price was cut from Tk 100 to Tk 90 on 16 March 2026, notebook sales at Tk 60 rose from 200 to 210 a week. Substitutes: when the price of a second-hand prescribed textbook fell by 20 per cent, weekly sales of the new copy fell from 50 to 44 (new copy price Tk 700). Snob effect: a numbered edition at Tk 2,400 sold out in three days, while the unnumbered Tk 1,200 edition took three weeks. Bandwagon effect: a title at Tk 450 sold 20 copies a week until it topped a public bestseller list, then 85 a week. Giffen good: no case in the shop; explain only in general terms. Question.

The term **Oligopoly** means a market with a few large sellers, each aware of the others' actions. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in pricing and the firm's goal, before moving to the next step.

The term **Profit maximisation** means choosing the output at which the gap between total revenue and total cost is greatest. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in pricing and the firm's goal, before moving to the next step.

The central question in pricing and the firm's goal is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in pricing and the firm's goal to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

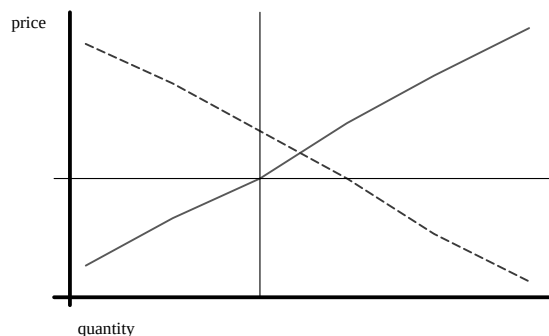


FIG. 11.1. a table-diagram of the market structures by number of sellers, product type and entry

REVIEW QUESTIONS

- 4.3 draw a curve from a new schedule, tell a movement from a shift in five events, say how a pen-pack price change affects notebook demand
 - 4.4 when the price of petrol rises, say what probably happens to demand for natural gas and why
 - 4.5 after the movement-versus-shift passage and after the exceptions passage
- PAUSE AND CHECK 4.P6 after the movement-versus-shift passage
- PAUSE AND CHECK 4.P7 the exceptions passage

Oligopoly. a market with a few large sellers, each aware of the others' actions

Oligopoly is the working idea for this part of the chapter.

Profit maximisation. choosing the output at which the gap between total revenue and total cost is greatest

Profit maximisation is the working idea for this part of the chapter.

Pricing and the Firm's Goal is understood by separating the question into small steps.

Pricing and the Firm's Goal is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

what is being measured or compared in monopoly, before moving to the next step.

The term **Pure competition** means a market with many sellers, an identical product and free entry, where buyers may lack full information or free movement. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in monopoly, before moving to the next step.

The term **Monopoly** means a market with one seller and no close substitute, so that seller sets the price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in monopoly, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in monopoly to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

11.4 | MONOPOLISTIC COMPETITION | NEW IDEA | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Pure monopoly** means a monopoly in which one seller supplies the whole market with a product that has no close substitute. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in monopolistic competition, before moving to the next step.

Pure competition. a market with many sellers, an identical product and free entry, where buyers may lack full information or free movement

Pure competition is the working idea for this part of the chapter.

Monopoly. a market with one seller and no close substitute, so that seller sets the price

Monopoly is the working idea for this part of the chapter.

Monopoly is understood by separating the question into small steps.

Monopoly is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what monopolistic competition means before applying it.

Pure monopoly. a monopoly in which one seller supplies the whole market with a product that has no close substitute

Pure monopoly is the working idea for this part of the chapter.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Teach how to measure how strongly buyers respond, with percentages, and what the answer means for revenue. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic); total revenue Tk 12,000 (60 x 200) to Tk 12,408 (66 x 188). Messier rung: midpoint method gives about 0.65. Second good: a Tk 300 atlas, price cut to Tk 270 raised weekly sales from 20 to 30: -10 per cent price, +50 per cent quantity, elasticity 5 in size (elastic); revenue Tk 6,000 to Tk 8,100. Income elasticity: a household income rose from Tk 30,000 to Tk 33,000 a month and its novels bought rose from 50 to 60 a year: +10 per cent and +20 per cent, elasticity 2. Cross elasticity: a rival shop raised its notebook price from Tk 50 to Tk 55 (+10 per cent) and Meghna and Sons sales of its own notebooks rose from 100 to 108 a week (+8 per cent), cross elasticity 0.8. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

5.1 | MEASURING RESPONSIVENESS | TECHNIQUE | 775

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Elasticity** means a measure of how strongly one quantity responds when another quantity changes. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in measuring responsiveness, before moving to the next step.

This section asks what measuring responsiveness means before applying it.

Elasticity. a measure of how strongly one quantity responds when another quantity changes
Elasticity is the working idea for this part of the chapter.

books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in types of market, before moving to the next step.

The term **Competitive market** means a market with many buyers and sellers, so no single one can set the price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in types of market, before moving to the next step.

The term **Industry** means all the firms that make the same or very similar goods. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in types of market, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

Before moving on, keep one question in view: what must remain true for the rule in types of market to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

11.2 | PERFECT COMPETITION | NEW IDEA | 775

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Price taker** means a firm that must accept the market price because it is too small to change it. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in perfect competition, before moving to the next step.

Competitive market. a market with many buyers and sellers, so no single one can set the price

Competitive market is the working idea for this part of the chapter.

Industry. all the firms that make the same or very similar goods

Industry is the working idea for this part of the chapter.

Types of Market is understood by separating the question into small steps.

Types of Market is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what perfect competition means before applying it.

Price taker. a firm that must accept the market price because it is too small to change it

Price taker is the working idea for this part of the chapter.

Elastic demand. demand for which the percentage fall in quantity is greater than the percentage rise in price

Elastic demand is the working idea for this part of the chapter.

Inelastic demand. demand for which the percentage fall in quantity is smaller than the percentage rise in price

Inelastic demand is the working idea for this part of the chapter.

Unit elastic demand. demand for which the percentage change in quantity equals the percentage change in price

Unit elastic demand is the working idea for this part of the chapter.

Forms of Price Elasticity is understood by separating the question into small steps.

Forms of Price Elasticity is understood by separating the question into small steps.

Forms of Price Elasticity is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

The term **Elastic demand** means demand for which the percentage fall in quantity is greater than the percentage rise in price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in forms of price elasticity, before moving to the next step.

The term **Inelastic demand** means demand for which the percentage fall in quantity is smaller than the percentage rise in price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in forms of price elasticity, before moving to the next step.

The term **Unit elastic demand** means demand for which the percentage change in quantity equals the percentage change in price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in forms of price elasticity, before moving to the next step.

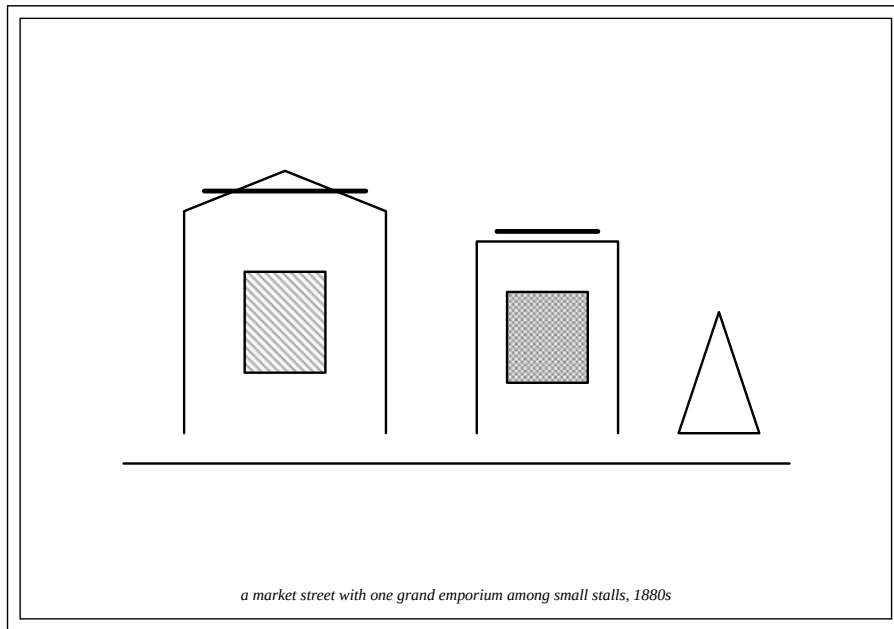
The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in forms of price elasticity to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *



CHAPTER XI

Market Structures and the Firm

What does this chapter let us explain that the earlier ideas could not explain alone?



Compare how firms behave when they face many rivals, one rival or none, and how each structure sets price and output. The chapter builds the idea in small stages, then applies it to the running case.

The examples keep the stated conditions visible so the reasoning can be checked.

This section asks what income and cross elasticity means before applying it.

Income elasticity of demand. the percentage change in quantity demanded divided by the percentage change in buyers' income

Income elasticity of demand is the working idea for this part of the chapter.

Cross elasticity of demand. the percentage change in quantity demanded of one good divided by the percentage change in another good's price

Cross elasticity of demand is the working idea for this part of the chapter.

Income and Cross Elasticity is understood by separating the question into small steps.

Income and Cross Elasticity is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

5.4 | INCOME AND CROSS ELASTICITY | TECHNIQUE | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Income elasticity of demand** means the percentage change in quantity demanded divided by the percentage change in buyers' income. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in income and cross elasticity, before moving to the next step.

The term **Cross elasticity of demand** means the percentage change in quantity demanded of one good divided by the percentage change in another good's price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in income and cross elasticity, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in income and cross elasticity is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in income and cross elasticity to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

workers and 8 machines Tk 14,400; 24 workers and 2 machines Tk 16,800; least cost Tk 12,000, where marginal product of labour is 30 and of a machine 60 (30 over 600 equals 60 over 1,200). Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Show what costs a firm counts, how they change with output, and how to reach a given output at the lowest cost. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

10.T1 complete the cost columns from new data, find where marginal cost crosses average cost, choose the cheapest input mix from three priced options

10.T2 explain why a family business should count its own unpaid time as a cost when judging success

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to Henry Bessemer's converter process, 1856, and the fall in the price of steel; search: Bessemer process 1856 steel price fall per ton. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

The chapter has built one connected chain of reasoning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

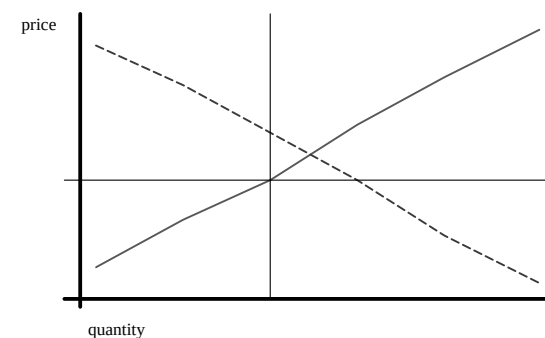


FIG. 5.3. total revenue before and after a price rise for an inelastic and an elastic good

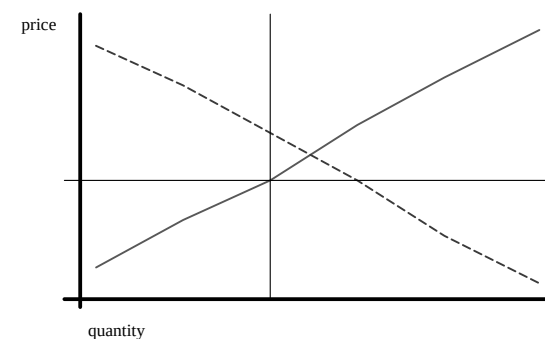


FIG. 5.4. a number line placing income elasticity values for necessities, comforts and luxuries

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic); total revenue Tk 12,000 (60 x 200) to Tk 12,408 (66 x 188). Messier rung: midpoint method gives about 0.65. Second good: a Tk 300 atlas, price cut to Tk 270 raised weekly sales from 20 to 30: -10 per cent price, +50 per cent quantity, elasticity 5 in size (elastic); revenue Tk 6,000 to Tk 8,100. Income elasticity: a household income rose from Tk 30,000 to Tk 33,000 a month and its novels bought rose from 50 to 60 a year: +10 per cent and +20 per cent, elasticity 2. Cross elasticity: a rival shop raised its notebook price from Tk 50 to Tk 55 (+10 per cent) and Meghna and Sons sales of its own notebooks rose from 100 to 108 a week (+8 per cent), cross elasticity 0.8. Question. What does the record show when the chapter's central rule is applied to these facts?

The central question in least-cost combination is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in least-cost combination to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

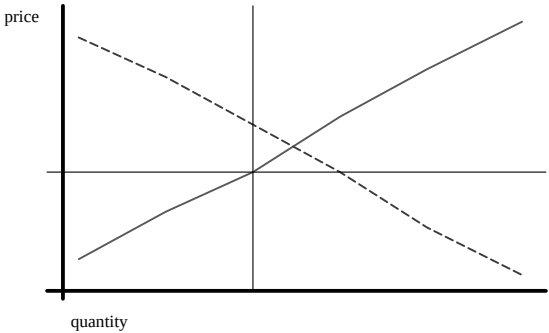


FIG. 10.1. fixed, variable and total cost curves on one diagram

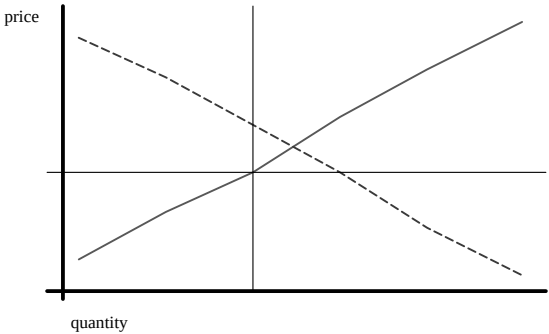


FIG. 10.2. average cost and marginal cost curves, with marginal cost cutting average cost at its lowest point

Least-Cost Combination is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

The chapter uses a small number of historical references to anchor its defining ideas.

SOURCES

Further reading and historical attribution should be checked against the original works and reliable archival or institutional sources. The chapter does not require the reader to accept a historical interpretation merely because it appears in a secondary account.

REVIEW QUESTIONS

- 5.3 compute three elasticities from new price-quantity pairs and name each form, say which goods have low elasticity and why, compute income and cross elasticity from new data
 - 5.4 a seller wants more revenue — decide from the elasticity whether to cut or raise the price
 - 5.5 after the formula passage and after the revenue passage
- PAUSE AND CHECK 5.P6 after the formula passage
- PAUSE AND CHECK 5.P7 the revenue passage

condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in short-run cost curves to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

10.3 | LONG-RUN COST CURVES | NEW IDEA | 225

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Prime cost** means the direct cost of materials and labour that go into a unit of output. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Outpu. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in long-run cost curves, before moving to the next step.

The term **Supplementary cost** means the cost of items apart from prime cost, such as rent and equipment, which remain even when output is nil. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Outpu. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in long-run cost curves, before moving to the next step.

The term **Long-run average cost curve** means a curve showing the lowest average cost of each output when every input can be changed. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Outpu. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in long-run cost curves, before moving to the next step.

The term **Isoquant** means a curve joining the input combinations that all make the same output. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost.

The section closes by restating the condition that controls the result.

This section asks what long-run cost curves means before applying it.

Prime cost. the direct cost of materials and labour that go into a unit of output
Prime cost is the working idea for this part of the chapter.

Supplementary cost. the cost of items apart from prime cost, such as rent and equipment, which remain even when output is nil
Supplementary cost is the working idea for this part of the chapter.

Long-run average cost curve. a curve showing the lowest average cost of each output when every input can be changed
Long-run average cost curve is the working idea for this part of the chapter.

Isoquant. a curve joining the input combinations that all make the same output
Isoquant is the working idea for this part of the chapter.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Show what a buyer gains from a purchase and how a buyer with a fixed budget picks the best bundle. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650; four buy and one does not. Surpluses: Tk 250, Tk 150, Tk 50, Tk 0, total Tk 450. Choice: a student's monthly spending plan is Tk 1,200 on prescribed books at Tk 150 each and stationery packs at Tk 100 each. Budget line bundles (books, packs): (0, 12), (2, 9), (4, 6), (6, 3), (8, 0). One indifference curve through (4, 6) also passes through (3, 8) and (6, 4); the slope between (3, 8) and (4, 6) is 2 packs per book, between (4, 6) and (6, 4) it is 1 pack per book. Best bundle: (4, 6), cost Tk 600 plus Tk 600 equals Tk 1,200; the price ratio 150 to 100 equals 1.5 packs per book. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

6.1 | CONSUMPTION AND CONSUMER SURPLUS | NEW IDEA | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Consumption** means the using up of goods and services to meet wants. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in consumption and consumer surplus, before moving to the next step.

This section asks what consumption and consumer surplus means before applying it.

Consumption. the using up of goods and services to meet wants
Consumption is the working idea for this part of the chapter.

what is being measured or compared in cost concepts, before moving to the next step.

The term **Explicit cost** means a cost paid out in money to others, such as rent or wages. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in cost concepts, before moving to the next step.

The term **Implicit cost** means the value of resources the owner supplies, such as own time, for which no payment is made. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in cost concepts, before moving to the next step.

The term **Fixed cost** means a cost that stays the same however much is produced in the short run. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in cost concepts, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in cost concepts to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

Explicit cost. a cost paid out in money to others, such as rent or wages

Explicit cost is the working idea for this part of the chapter.

Implicit cost. the value of resources the owner supplies, such as own time, for which no payment is made

Implicit cost is the working idea for this part of the chapter.

Fixed cost. a cost that stays the same however much is produced in the short run

Fixed cost is the working idea for this part of the chapter.

Cost Concepts is understood by separating the question into small steps.

Cost Concepts is understood by separating the question into small steps.

Cost Concepts is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

Indifference map. a set of indifference curves, with curves farther from the origin showing more satisfaction

Indifference map is the working idea for this part of the chapter.

Marginal rate of substitution. the amount of one good a consumer will give up to gain one more unit of the other

Marginal rate of substitution is the working idea for this part of the chapter.

Indifference Curves and the Budget Line is understood by separating the question into small steps.

Indifference Curves and the Budget Line is understood by separating the question into small steps.

Indifference Curves and the Budget Line is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what consumer equilibrium means before applying it.

Budget line. a line showing every bundle of two goods a consumer can exactly afford with a given income

Budget line is the working idea for this part of the chapter.

The term **Indifference map** means a set of indifference curves, with curves farther from the origin showing more satisfaction. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in indifference curves and the budget line, before moving to the next step.

The term **Marginal rate of substitution** means the amount of one good a consumer will give up to gain one more unit of the other. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in indifference curves and the budget line, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

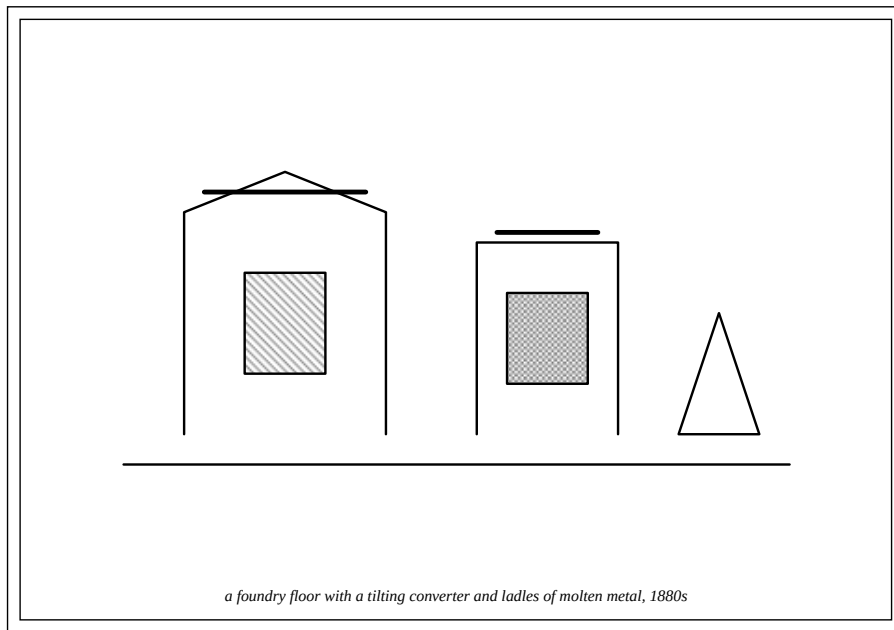
Before moving on, keep one question in view: what must remain true for the rule in indifference curves and the budget line to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

6.3 | CONSUMER EQUILIBRIUM | TECHNIQUE | 650

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Budget line** means a line showing every bundle of two goods a consumer can exactly afford with a given income. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk



CHAPTER X

Costs of Production

What does this chapter let us explain that the earlier ideas could not explain alone?



Show what costs a firm counts, how they change with output, and how to reach a given output at the lowest cost. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

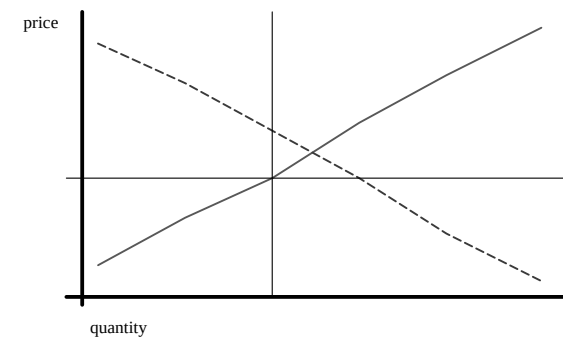


FIG. 6.2. a family of indifference curves, convex to the origin

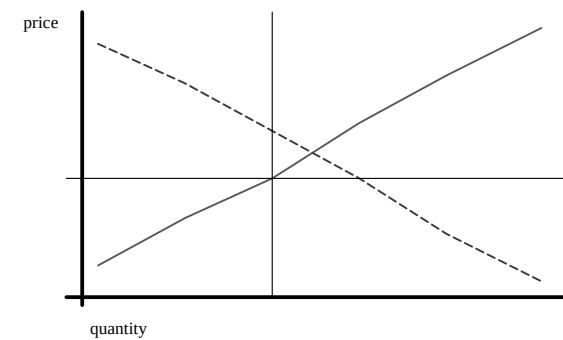


FIG. 6.3. a budget line drawn from a spending plan of Tk 1,200

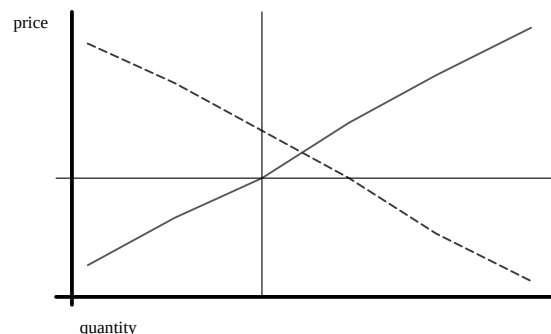


FIG. 9.4. three paths of output as scale doubles: more than double, double, less than double

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12, 12, 11, 9.6, 8. Long run, scale tables (binds a day): one machine and 2 workers, 40; two machines and 4 workers, 90; four machines and 8 workers, 180; eight machines and 16 workers, 300. Types: first doubling gives 2.25 times output (increasing), second doubling gives 2.0 times (constant), third gives about 1.67 times (decreasing). Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Explain how inputs become output, why adding one input yields less and less, and what changes when all inputs grow. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been

The worked case applies the chapter without adding facts to the record.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

The chapter has built one connected chain of reasoning.

rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650; four buy and one does not. Surpluses: Tk 250, Tk 150, Tk 50, Tk 0, total Tk 450. Choice: a student's monthly spending plan is Tk 1,200 on prescribed books at Tk 150 each and stationery packs at Tk 100 each. Budget line bundles (books, packs): (0, 12), (2, 9), (4, 6), (6, 3), (8, 0). One indifference curve through (4, 6) also passes through (3, 8) and (6, 4); the slope between (3, 8) and (4, 6) is 2 packs per book, between (4, 6) and (6, 4) it is 1 pack per book. Best bundle: (4, 6), cost Tk 600 plus Tk 600 equals Tk 1,200; the price ratio 150 to 100 equals 1.5 packs per book. Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

The same method remains useful when the surrounding details change.

The same method remains useful when the surrounding details change.

The worked case applies the chapter without adding facts to the record.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

The term **Decreasing returns to scale** means output rises by a smaller proportion than inputs. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in returns to scale, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in returns to scale is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in returns to scale to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

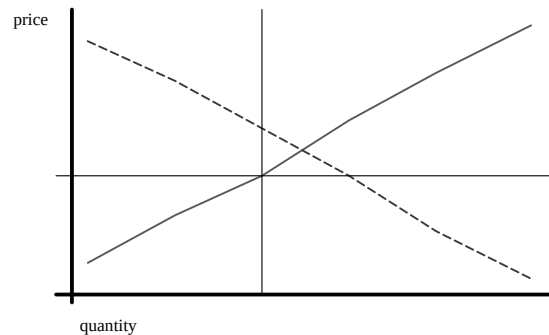


FIG. 9.1. a total product curve rising then flattening and a marginal product curve peaking then falling

Decreasing returns to scale.

output rises by a smaller proportion than inputs

Decreasing returns to scale is the working idea for this part of the chapter.

Returns to Scale is understood by separating the question into small steps.

Returns to Scale is understood by separating the question into small steps.

Returns to Scale is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

* * *

9.3 | DIMINISHING RETURNS | DEFINING THEORY | 775

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Total product** means the whole output from a given amount of inputs. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in diminishing returns, before moving to the next step.

The term **Average product** means total product divided by the number of units of the variable input. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in diminishing returns, before moving to the next step.

The term **Marginal product** means the extra output from adding one more unit of the variable input. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in diminishing returns, before moving to the next step.

The term **Law of diminishing returns** means when more of one input is added to fixed inputs, the extra output eventually falls. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in diminishing returns, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

This section asks what diminishing returns means before applying it.

Total product. the whole output from a given amount of inputs

Total product is the working idea for this part of the chapter.

Average product. total product divided by the number of units of the variable input

Average product is the working idea for this part of the chapter.

Marginal product. the extra output from adding one more unit of the variable input

Marginal product is the working idea for this part of the chapter.

Law of diminishing returns. when more of one input is added to fixed inputs, the extra output eventually falls

Law of diminishing returns is the working idea for this part of the chapter.

Diminishing Returns is understood by separating the question into small steps.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Build the supply idea from a price table to a curve and measure how strongly sellers respond to price. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300; Tk 60, 350. Elasticity from Tk 50 to Tk 55: +10 per cent price, +20 per cent quantity, elasticity 2.0 (elastic). Shift: on 1 April 2026 the printer's paper cost rose and it offered 50 fewer books at every price: 100, 150, 200, 250, 300. Exceptional curve 1: a rare first edition, only one copy exists, supply unchanged at any price. Exceptional curve 2: a part-time binder offers 20 hours a week at Tk 150 an hour, 26 hours at Tk 200, 24 hours at Tk 260. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

7.1 | THE LAW OF SUPPLY | DEFINING THEORY | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Supply** means the quantity of a good sellers are willing and able to offer at each possible price, in a given period. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of supply, before moving to the next step.

The term **Quantity supplied** means the amount sellers plan to offer at one particular price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader

This section asks what the law of supply means before applying it.

Supply. the quantity of a good sellers are willing and able to offer at each possible price, in a given period

Supply is the working idea for this part of the chapter.

Quantity supplied. the amount sellers plan to offer at one particular price

Quantity supplied is the working idea for this part of the chapter.

12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the production function, before moving to the next step.

The term **Output** means the goods or services produced. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the production function, before moving to the next step.

The term **Production function** means a rule or table linking the quantities of inputs used to the greatest output they can make. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the production function, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in the production function to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

9.2 | SHORT RUN AND LONG RUN | FRAMEWORK | 225

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

Output, the goods or services produced

Output is the working idea for this part of the chapter.

Production function, a rule or table linking the quantities of inputs used to the greatest output they can make

Production function is the working idea for this part of the chapter.

The Production Function is understood by separating the question into small steps.

The Production Function is understood by separating the question into small steps.

The Production Function is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what short run and long run means before applying it.

Determinants of supply, the things other than the good's own price that change how much sellers offer

Determinants of supply is the working idea for this part of the chapter.

What Moves the Supply Curve is understood by separating the question into small steps.

What Moves the Supply Curve is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what exceptions to the law means before applying it.

Backward-bending supply

curve, a curve showing that beyond some pay level a worker offers fewer hours as pay rises further

Backward-bending supply curve is the working idea for this part of the chapter.

Price elasticity of supply, the percentage change in quantity supplied divided by the percentage change in price

Price elasticity of supply is the working idea for this part of the chapter.

measured or compared in what moves the supply curve, before moving to the next step.

The term **Determinants of supply** means the things other than the good's own price that change how much sellers offer. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what moves the supply curve, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in what moves the supply curve to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

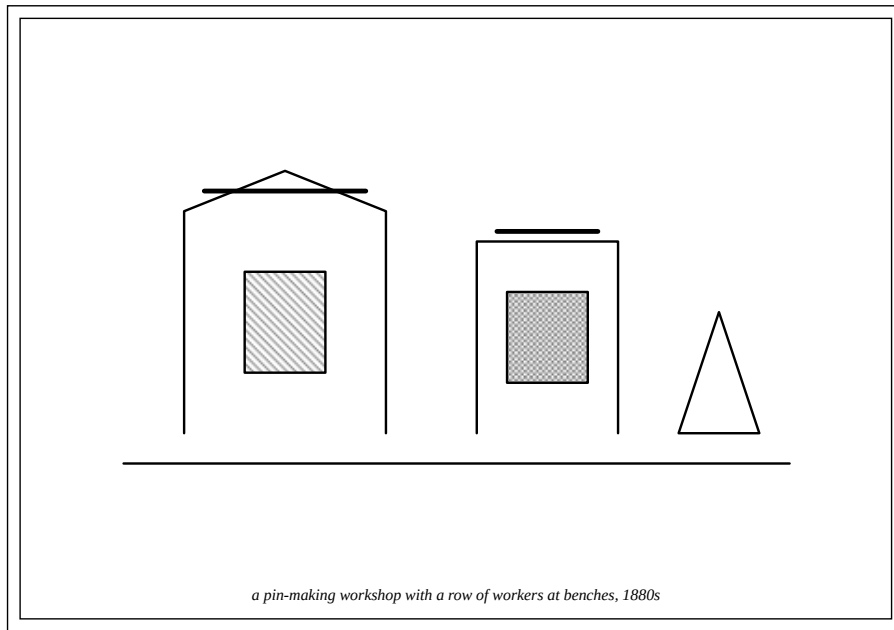
* * *

7.3 | EXCEPTIONS TO THE LAW | FRAMEWORK | 225

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Backward-bending supply curve** means a curve showing that beyond some pay level a worker offers fewer hours as pay rises further. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in exceptions to the law, before moving to the next step.

The term **Price elasticity of supply** means the percentage change in quantity supplied divided by the percentage change in price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is



CHAPTER IX

Production

What does this chapter let us explain that the earlier ideas could not explain alone?



Explain how inputs become output, why adding one input yields less and less, and what changes when all inputs grow. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

Elasticity of Supply is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in elasticity of supply to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

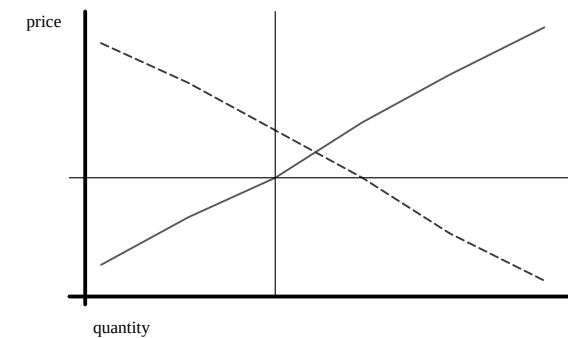


FIG. 7.1. a supply curve drawn from the printer's schedule

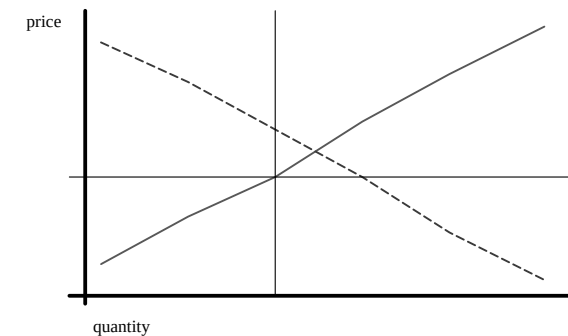


FIG. 7.2. a shift of the supply curve with its causes labelled

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Bring demand and supply together to find a price, show why a market is pulled toward it, and measure who gains. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

8.T1 find equilibrium from new equations, describe the adjustment from a price above or below it, trace the effect of a shift

8.T2 a government sets a price ceiling below equilibrium for a staple — describe who gains and who loses

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to Diocletian's Edict on Maximum Prices, AD 301; search: Diocletian Edict on Maximum Prices 301 shortages. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

The chapter has built one connected chain of reasoning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300; Tk 60, 350. Elasticity from Tk 50 to Tk 55: +10 per cent price, +20 per cent quantity, elasticity 2.0 (elastic). Shift: on 1 April 2026 the printer's paper cost rose and it offered 50 fewer books at every price: 100, 150, 200, 250, 300. Exceptional curve 1: a rare first edition, only one copy exists, supply unchanged at any price. Exceptional curve 2: a part-time binder offers 20 hours a week at Tk 150 an hour, 26 hours at Tk 200, 24 hours at Tk 260. Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Build the supply idea from a price table to a curve and measure how strongly sellers respond to price. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

The same method remains useful when the surrounding details change.

The worked case applies the chapter without adding facts to the record.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

The chapter has built one connected chain of reasoning.

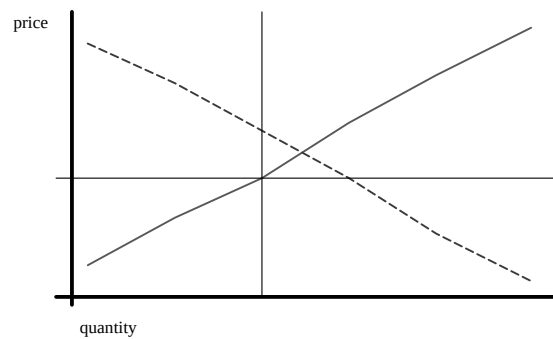


FIG. 8.2. a ceiling below and a floor above equilibrium with the shortage and surplus gaps marked

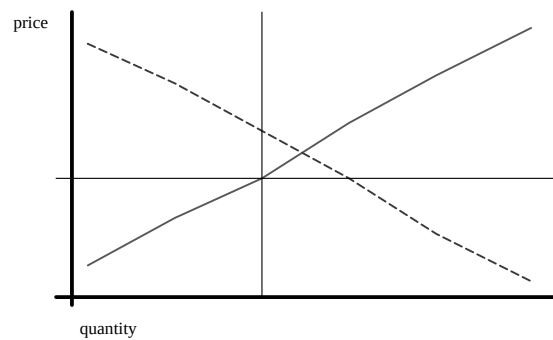


FIG. 8.3. two panels: a supply shift left and a demand shift right with the new equilibria

70: In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in shifts and new equilibria, before moving to the next step.

The term **Producer surplus** means the gap between the price a seller receives and the lowest price the seller would accept. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = Tk\ 60$, $Q = 900$. Ceiling at Tk 50: $Q_d\ 1,050$, $Q_s\ 650$, excess demand 400. Floor at Tk 70: In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in shifts and new equilibria, before moving to the next step.

The term **Total surplus** means consumer surplus plus producer surplus, a measure of the gain from trade. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = Tk\ 60$, $Q = 900$. Ceiling at Tk 50: $Q_d\ 1,050$, $Q_s\ 650$, excess demand 400. Floor at Tk 70: In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in shifts and new equilibria, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in shifts and new equilibria is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in shifts and new equilibria to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

8.4 | CONSUMER AND PRODUCER SURPLUS IN PRACTICE | NEW IDEA | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the

Producer surplus, the gap between the price a seller receives and the lowest price the seller would accept

Producer surplus is the working idea for this part of the chapter.

Total surplus, consumer surplus plus producer surplus, a measure of the gain from trade

Total surplus is the working idea for this part of the chapter.

Shifts and New Equilibria is understood by separating the question into small steps.

Shifts and New Equilibria is understood by separating the question into small steps.

Shifts and New Equilibria is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what consumer and producer surplus in practice means before applying it.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

This section asks what markets and market equilibrium means before applying it.

Equilibrium, a state in which nothing pushes the price or quantity to change

Equilibrium is the working idea for this part of the chapter.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Bring demand and supply together to find a price, show why a market is pulled toward it, and measure who gains. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = Tk\ 60$, $Q = 900$. Ceiling at Tk 50: $Q_d\ 1,050$, $Q_s\ 650$, excess demand 400. Floor at Tk 70: $Q_d\ 750$, $Q_s\ 1,150$, excess supply 400. Supply shift: paper cost rises, supply becomes $Q_s = 25P - 760$; new equilibrium $P = Tk\ 64$, $Q = 840$. Demand intercept price is Tk 120; supply intercept price is Tk 24 before the shift. Consumer surplus at equilibrium: $0.5 \times (120 - 60) \times 900 = Tk\ 27,000$ a week; producer surplus: $0.5 \times (60 - 24) \times 900 = Tk\ 16,200$ a week. After the shift: consumer surplus $0.5 \times (120 - 64) \times 840 = Tk\ 23,520$; producer surplus $0.5 \times (64 - 30.4) \times 840 = Tk\ 14,112$. Substitute-goods link: when the price of petrol rises, demand for natural gas shifts right (kept as a general example, not shop data). The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

8.1 | MARKETS AND MARKET EQUILIBRIUM | FRAMEWORK | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Equilibrium** means a state in which nothing pushes the price or quantity to change. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = Tk\ 60$, $Q = 900$. Ceiling at Tk 50: $Q_d\ 1,050$, $Q_s\ 650$, excess demand 400. Floor at Tk 70: In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in markets and market equilibrium, before moving to the next step.